

AnyBlox

Liam Sanft

liam.sanft@mailbox.tu-dresden.de

TU Dresden

Germany, Dresden

ABSTRACT

Here comes the abstract.

1 INTRODUCTION

For decades, database systems operated as tightly coupled units in which storage format, processing logic, and data access were controlled by a single system. External systems submitted queries and received results, while everything in between remained internal concerns. With full control over the physical layout, systems could optimize both storage and processing of data freely within their own boundaries. As long as data resided within a single system, this model was sufficient [6].

However, the volume and variety of data collected has grown drastically. Data arrives continuously from a growing number of sources, in formats tailored to their respective domains rather than to any particular database system. As a result, the database landscape has evolved from a single system controlling all aspects of data management toward a heterogeneous, modular ecosystem of specialized systems, each handling a distinct part of the data pipeline and accessing a shared storage layer. Open file formats such as Apache Parquet [4] and ORC [3], stored in data lakes [1], have gained wide adoption in this context. Beyond these, data scientists increasingly need to analyze data in domain-specific encodings from fields such as high-energy physics and bioinformatics [5].

Independently, research in data compression and encoding continues to produce formats that outperform established standards

in storage efficiency and query throughput. These advances rarely reach production systems, however, because each system must independently implement support for every format it wishes to read [5].

Both trends converge on the same structural problem. Codd’s principle of physical data independence [2], which assumes that a system controls its own storage format and shields applications from its physical details, no longer holds in a shared storage layer. In a shared storage layer, this assumption no longer holds. The format is determined by whoever produced the data. Every system that wants to read a given format must implement its own decoder. With N systems and M formats in active use, this results in $N \times M$ independent implementations to develop and maintain [5].

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